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Dear Editors,

I am pleased to submit my manuscript, “**Conformal Surprisal Fusion for Detecting Prompt-Injection Compromise in LLM Agent Traces**,” for consideration as a research article in the *Journal of Information Security and Applications*.

Large language model agents that call external tools now read email, browse the web and move money on a user’s behalf. Because they ingest untrusted content and fold it into the context that drives their next action, they are exposed to indirect prompt injection. Runtime monitors that watch only the black-box execution trace need no model access and no retraining, but the literature has converged almost entirely on one kind of evidence: whether untrusted text *looks like an instruction*. This paper asks what else the trace affords, how heterogeneous evidence should be combined, and where the information in a trace runs out.

The work makes five contributions. First, it introduces **SENTRY-Fuse**, a monitor that converts heterogeneous trace signals—one scoring the observation stream, one auditing the action stream—into conformal tail surprisals against a benign calibration split and fuses them by addition. The construction is scale-free, non-decreasing in every component and free of tuned signal weights, and the paper proves that the natural alternative of subtracting the benign training maximum is many-to-one and, for any attack class lying below that maximum, caps AUROC at one half. That is not hypothetical: an earlier, unpublished draft of this work did exactly that, at a cost of 0.25 AUROC and a confidently wrong explanation of the resulting failure. Second, action-side evidence is shown to be *complementary* rather than redundant, and I am careful about how strongly. Over 90 genuinely compromised trajectories, adding one action-side audit to the instruction signal raises AUROC from 0.937 to 0.953 and recall from 0.709 to 0.898. Resampling trajectories, the recall gain is +0.1706 [+0.0033, +0.3533] and the AUROC gain +0.0163 [−0.0006, +0.0351]—the latter interval only marginally excluding no effect, which a corpus of this size cannot settle, and I say so in the paper rather than assert significance. The effect I do defend is different in kind: the single signal returns *zero* recall in 2 of 10 calibration draws, because a saturating judge and a discrete operating point put the threshold at the ceiling, and fusion removes that failure entirely. A third signal I implemented, sink provenance, does not reach significance and is reported as an ablation rather than folded into the method.

Third, I measure where trace evidence lives and draw a benchmark-design conclusion: AgentDojo’s observation channel is close to saturated, because it delivers payloads through tool outputs a successful injection must have read, so corpora of this shape cannot contain a compromise invisible to a trace monitor and cannot test the hard case. I explicitly decline to convert this into an information-theoretic ceiling, and explain in the paper why an earlier draft was wrong to do so—the quantity is not identified, and at the strictest criterion the bound it yields is violated by my own measured true-positive rate.

Fourth, I document a label-polarity trap in the standard AgentDojo harness. Its per-trajectory **security** flag records whether the *injection* succeeded, so reading it as “the system was secure” inverts the compromised and resisted classes. This inverted my own results for a substantial period, and the paper gives the in-trace ground-truth procedure that settles it.

Fifth, because a comparison between one’s own signal subsets is a self-ablation, I re-implemented **Task Shield**—the closest prior work to my action-side signal—and ran it on my own traces, under my own splits, with the same judge model, so that the comparison isolates the method rather than the defender. The fusion ranks better (0.953 against 0.764 pooled AUROC). Task Shield’s published rule catches more compromises than mine does (0.933 against 0.898) but flags 39% of benign trajectories, and because its score takes only seven distinct values with 39% of benign traces tied at the floor, it has no operating point at any lower false-alarm rate. I state plainly what this does not establish: it is not a verdict on Task Shield as a *defence*, since it blocks the calls it flags and I only raise a flag, and one of its two checking layers has no counterpart in my traces.

I should also be explicit that the paper does *not* claim to beat the state of the art. The strongest reported trace-only figure (95.75% true positives at 3.66% false positives) comes from a preprint version that later versions withdraw and was measured on a different corpus with a different agent; I cite it as a reference point for the regime, not as a baseline I outperform.

I believe the work suits the *Journal of Information Security and Applications* in particular. It is a security paper first, not a machine-learning paper: it defines a threat model, builds a runtime monitor, evaluates it against real

attacks on real execution traces, and reports its operating limits rather than only its successes. The detector itself is a conformal hypothesis test over trace-derived scores; the language model appears only as a scorer, and the contribution is the calibration and fusion argument around it. This places the work alongside the journal's recent coverage of the security of learned systems and of learning-based detection under a stated false-alarm budget.

The paper offers two things of direct use to others beyond the monitor: a measurement showing that the benchmark the field relies on cannot exhibit the hardest case, and a methodological warning about how results on that benchmark should be read. Its negative results are reported in full—five components I implemented and rejected, and the finding that a hand-picked threshold outperformed the anytime-valid machinery from which the project began.

All code, the trajectory corpora, the cached model-judge scores and the analysis scripts that regenerate every number, table and figure are publicly released at

<https://github.com/vinhqdang/SENTRY-Anytime-Valid-E-Process-Monitoring-of-LLM-Agent-Action-Streams> and the evaluation reproduces offline, via a single `./reproduce.sh`, without access to any model API.

I confirm that this manuscript is original, has not been published previously and is not under consideration elsewhere. There are no competing interests and no external funding was received; as sole author I am responsible for all aspects of the work. Thank you for your consideration, and I look forward to the reviewers' comments.

Sincerely,

Quang-Vinh Dang